Camera Ready Summary

Conference Name

12th IEEE International Women in Engineering (WIE) Conference on Electrical and Computer Engineering 2026

Paper ID

518

Paper Title

OrdinalFed: Federated Learning with Ordinal-Aware Aggregation for Diabetic Retinopathy Grading

Abstract

Diabetic Retinopathy (DR) is a leading cause of preventable blindness. With federated learning (FL), hospitals can train one DR grading model together without sharing retinal images. When hospitals see very different patients, however, FedAvg can collapse. In this paper, OrdinalFed is studied. It has two parts. The first is an OrdinalLoss that penalises a wrong grade by its squared distance from the true grade. The second is a QWK-weighted aggregation: the server scores each client model by its Quadratic Weighted Kappa (QWK) on 50 held-out images, and over the rounds it moves the aggregation weights from image counts towards these scores. All experiments on the DDR dataset were repeated with three seeds and scored on the held-out test split. Under extreme label skew (Dirichlet parameter 0.1), FedAvg collapsed in every seed, with a worst-round test QWK of 0.378. OrdinalFed raised the worst round to 0.514, with no change at the hospitals and about 1.2 s of extra server time per round. The ablations show where this gain comes from: small hospitals get more weight. Equal client weights did even better (0.630), while the QWK scores and the OrdinalLoss added no measurable benefit. FedProx was the most stable method (0.641), and OrdinalFed with FedProx's proximal term came close to it (0.627). On the IID and realistic multi-site partitions, all methods were within 0.01 QWK of each other. On the external APTOS 2019 dataset, all models scored 0.597–0.658 QWK, close to the centralised model (0.641).

Keywords—Diabetic Retinopathy, Federated Learning, Non

IID, Ordinal Loss, QWK-Weighted Aggregation, EfficientNet-B3, Medical Imaging

Authors

Md Tamim Hossain - tamim0072276@gmail.com

Najifa Raksanda - najifa.raksandaa@gmail.com

M. M.Tamim Sharif - tamimsharif2181@gmail.com

Aria Nawshin Ashka - thearianawshin@gmail.com

Camera Ready Files

518.pdf (368.4 Kb, 2026-10-17 23:09:00 GMT+6)

518_express.pdf (80.8 Kb, 2026-10-17 23:20:57 GMT+6)

518_payment.pdf (51.6 Kb, 2026-10-17 23:20:57 GMT+6)

518_copyright.pdf (34.2 Kb, 2026-10-17 23:20:57 GMT+6)

518_student.pdf (205.2 Kb, 2026-10-17 23:20:57 GMT+6)

Camera Ready Questions Response

1. Camera Ready Submission Guideline

Agreement accepted

2. First page header info

Agreement accepted

3. First page footer info

Agreement accepted

4. Page Number Removal

Agreement accepted

5. IEEE e-copyright form

Agreement accepted

6. File Naming Confirmation

Agreement accepted

7. Plagiarism

Agreement accepted

8. IEEE Xplore rights

Agreement accepted

9. Registration

Agreement accepted

10. Presentation File (video and ppt slides)

NA

11. Mode of Presentation(Physical/Online)

Physical

12. IEEE PDF Express compatible checking

Agreement accepted

13. Online Presentation Request

Not Applicable for me (I will present in-person)

14. Conference Kits for Online mode of presentation

Agreement accepted

15. Presenting Author Name

Md Tamim Hossain

16. Affiliation of the presenting author

Department of Computer Science and Engineering, Military Institute of Science and Technology (MIST), Mirpur Cantonment, Dhaka-1216, Bangladesh

17. Emergency Contact Number

01916507986

18. Female Author Confirmation

Agreement accepted

19. Name of at least one Female Author

Agreement accepted

20. Whether the Female Author is IEEE WIE member

No